

Final Assessment Test (FAT) - May 2024

Programme	B.Tech.	Semester	WINTER SEMESTER 2023 - 24
Course Title	COMPUTER AIDED DESIGN AND FINITE ELEMENT ANALYSIS	Course Code	BMEE306L
Faculty Name	Prof. Lakshmi Pathi Jakkamputi	Slot	B1+TB1
		Class Nbr	CH2023240501135
Time	3 Hours	Max. Marks	100
General Instructions:			
Write only Register Number in the Question Paper where space is provided (right-side at the top) & do not write any other details.			

Answer any 5 questions (5 X 20 Marks = 100 Marks)

- (a) What are the different types of Input and Output devices used in CAD system? Explain with the help of block diagram, the hardware structure of CAD system and the function of design workstation. (10 marks) [20]

(b) Find out the new transformed position of a triangle ABC having vertices A (3,-1), B (4, 1), C (2, 1) is rotated by 60° in clockwise direction and then reflected about X-axis and followed by successive scaling with scaling factor of 2 unit in X and Y direction. (10 marks)
- (a) What is the basic function for a Bezier curve and its important applications? (12 marks) [20]

(b) What are B-spline curve? Discuss its important properties. (8 marks)
- (a) What are the graphic standards? Explain IGES file structure with examples. (8 marks) [20]

(b) List down the various solid model construction techniques available in CAD. Explain any one of them in detail. (12 marks)
- A bar of uniform cross-section is clamped at one end and left free at the other end and it is subjected to a load P as shown in Figure.1. Determine the displacement and stress in a bar by using three terms polynomial. Compare with exact solutions. [20]

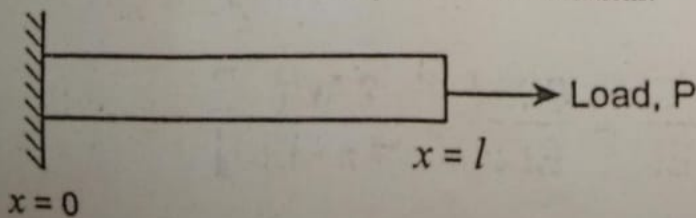


Figure.1 A bar of uniform cross-section

05. A bar is subjected to an axial load of $P=600$ kN as shown in figure.2. Divide the domain into two elements. Determine the following: (a) Displacement at each node. (b) Stresses in each element, and (c) Reactions at each node point. Assume, $A=250$ mm² and $E=2 \times 10^5$ N/mm². (5+6+9 Marks) [20]

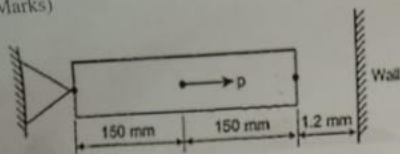


Figure.2

06. Derive the shape function and stiffness matrix for one dimensional quadratic 2 noded bar element using natural coordinate system. [20]
07. For the plane stress element shown in figure.3, the nodal displacements are $u_1=2.0$ mm, $u_2=0.5$ mm, $u_3=3.0$ mm, $v_1=1.0$ mm, $v_2=0.0$ mm, $v_3=1.0$ mm. Determine the element stresses σ_x , σ_y and τ_{xy} . Also, determine the principal stresses (σ_1 , σ_2 and τ_{max}) and principal plane (θ_p). Assume $E=210$ GPa, $\nu=0.25$ and $t=10$ mm. All co-ordinates are in millimetre. [20]

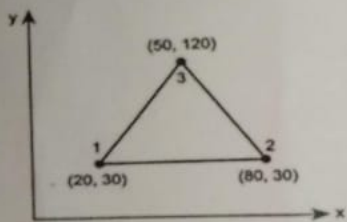


Figure.3

08. For the one-dimensional bar shown in Figure.4, determine the first two natural frequencies of longitudinal vibration using two elements of equal length. Take $E=2 \times 10^5$ N/mm², 0.8×10^{-4} N/mm³, and $L=400$ mm. [20]

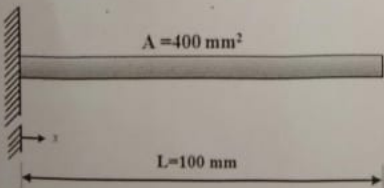


Figure. 4